

SCALAR CLASSES IN CRITICAL HALL MULTIPLICATION

1.1 THE LOWEST COEFFICIENT AT A SCALAR TRIPLE

At a scalar triple, every subspace is invariant. Its Hall coefficient still depends on the ambient cubic potential. Complete flags produce a distinguished class in this coefficient, including at the scalar point. The class determines the lowest-degree images for the potentials of ranks three, four, and five.

All spaces are complex algebraic varieties with their analytic topology. Coefficients are rational. For $1 \leq n \leq 5$, put

$$V_n = \text{Mat}_n(\mathbb{C})^3, \quad f_n(A, B, C) = \text{Tr}(A[B, C]), \quad Z_n = \{[A, B] = [B, C] = [C, A] = 0\}.$$

Differentiation using the trace pairing gives $Z_n = \text{Crit}(f_n)$. Write $\mathfrak{X}_n = [V_n/\text{GL}_n]$ and $\mathfrak{M}_n = [Z_n/\text{GL}_n]$. Constructible complexes on these quotient stacks retain the conjugation action and its descent data. Their ordinary atlas pullbacks have no additional shift. The coefficient is

$$\mathcal{P}_n = \Phi_{f_n} \mathbb{Q}_{\mathfrak{X}_n}[2n^2] \big|_{\mathfrak{M}_n}, \quad \Phi_f K = \text{Cone}(i^* K \longrightarrow \psi_f K)[-1],$$

where $i : f^{-1}(0) \hookrightarrow V$ and ψ_f denotes nearby cycles. This normalization agrees with Massey[3, pp. 1–2]. Thus $H^j(K[a]) = H^{j+a}(K)$ and $\Phi_0 K = K$. The tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy$. No completion or change of coefficients is used.

For $s = (aI_n, bI_n, cI_n)$, write $P_{n,s}$ for the ordinary atlas stalk of $\mathcal{P}_n$. Translation by scalar matrices preserves f_n , so it suffices to compute $P_{n,0}$. Define the commuting-pair variety and its complement by

$$C_n = \{(A, C) \in \text{Mat}_n(\mathbb{C})^2 : [C, A] = 0\}, \quad U_n = \text{Mat}_n(\mathbb{C})^2 \setminus C_n.$$

Borel–Moore homology below uses locally finite rational chains and the complex orientation of each smooth stratum.

PROPOSITION 1.1.1 (*Scalar stalk and commuting pairs*). For $2 \leq n \leq 5$ and every integer j , there are isomorphisms

$$H^j(P_{n,s}) \simeq \tilde{H}^{j+2n^2-1}(U_n; \mathbb{Q}) \simeq H_{2n^2-j}^{\text{BM}}(C_n; \mathbb{Q}). \quad (1.1)$$

In particular,

$$H^j(P_{n,s}) = 0 \quad (j < -2n), \quad H^{-2n}(P_{n,s}) \simeq \mathbb{Q}.$$

The same last two assertions hold for $n = 1$, where $P_{1,s} = \mathbb{Q}[2]$. Potential monodromy acts as the identity on the entire scalar stalk complex for every rank under consideration.

Proof. Use the positive real ray in the potential plane. By Massey[3, §1, p. 2], the normalized vanishing-cycle stalk is computed by cochains with support in $P = \{\text{Re } f_n \leq 0\}$, restricted to the origin. The fibre

convention is the relative-cochain convention $\text{Fib}(C^*(B) \rightarrow C^*(B \setminus P))$. Both P and its complement are invariant under positive real scaling. The radial map

$$x \mapsto \frac{x}{1 + \|x\|/\delta}$$

maps V_n into the ball of radius δ . Interpolation of the denominator gives homotopy inverses for the inclusions of these pairs. Thus their relative cohomology computes the scalar stalk as δ decreases to zero.

Let $F_n = f_n^{-1}(1)$ and $H = \{z \in \mathbb{C} : \text{Re } z > 0\}$. Choose the cubic root on H that is positive on the positive real axis. Homogeneity gives mutually inverse homeomorphisms

$$V_n \setminus P \longrightarrow F_n \times H, \quad x \mapsto (f_n(x)^{-1/3}x, f_n(x)), \quad (y, z) \mapsto z^{1/3}y.$$

Since V_n and H are contractible, the support complex has cohomology $\tilde{H}^{k-1}(F_n; \mathbb{Q})$ in degree k . This argument fixes the local-to-global comparison at the scalar point.

The cyclic trace identity writes the actual potential as $f_n = \text{Tr}(B[C, A])$. Projection onto (A, C) gives

$$F_n \longrightarrow U_n, \quad (A, B, C) \mapsto (A, C),$$

an affine bundle of complex rank $n^2 - 1$. Indeed, its fibre is the affine hyperplane $\text{Tr}(BD) = 1$, where $D = [C, A] \neq 0$. The trace pairing is nondegenerate. A continuous section is

$$B_0(A, C) = \frac{D^*}{\text{Tr}(D^*D)},$$

where D^* is the conjugate transpose. The affine homotopy from B to B_0 stays in this hyperplane. Therefore projection induces the first isomorphism in (1.1), after the shift $[2n^2]$. Projection is conjugation equivariant. The chosen section only supplies an ordinary topological homotopy inverse.

The real dimension of the pair space is $4n^2$. Its only nonzero Borel–Moore group is $H_{4n^2}^{\text{BM}} = \mathbb{Q}$. The closed-open exact sequence for C_n and U_n , followed by Poincaré duality on the smooth open set U_n , gives

$$\tilde{H}^k(U_n; \mathbb{Q}) \simeq H_{4n^2-k-1}^{\text{BM}}(C_n; \mathbb{Q}).$$

For $k = 0$, the reduced group removes the ambient fundamental class. The complement of a proper complex subvariety in this affine space is connected: a generic complex line joins each point to a common generic intermediate point while avoiding finitely many intersections. The remaining nonpositive degrees vanish. Substitution gives the second isomorphism in (1.1).

For completeness, the top dimension of C_n follows directly from Jordan forms. Fix a Jordan type for A with r distinct eigenvalues and a centralizer of dimension c . Its matrix stratum has dimension $n^2 - c + r$. The matrices C commuting with A form a vector bundle of rank c on this stratum. The corresponding pair stratum has dimension $n^2 + r$. There are finitely many Jordan types. The unique stratum with $r = n$ has dimension $n^2 + n$ and is irreducible: ordered eigenvalues and diagonal entries of C parameterize it, together with a change of basis. The finite permutation quotient preserves irreducibility. Every other stratum has dimension at most $n^2 + n - 1$. Consequently the top Borel–Moore group is

$$H_{2n^2+2n}^{\text{BM}}(C_n; \mathbb{Q}) = \mathbb{Q}, \quad H_m^{\text{BM}}(C_n; \mathbb{Q}) = 0 \quad (m > 2n^2 + 2n).$$

The top generator is the complex fundamental class of the closure of that stratum. No assertion about lower-dimensional components is needed. Equation (1.1) gives the degree bounds.

Finally, at the scalar origin the maps

$$(A, B, C) \mapsto (A, e^{i\theta} B, C), \quad 0 \leq \theta \leq 2\pi,$$

preserve every centered Euclidean ball and multiply f_n by $e^{i\theta}$. They trivialize the Milnor fibres over each circle in the potential plane. At 2π the map is the identity. These trivializations commute with restriction to smaller balls, with the constant-to-nearby-cycle map, and with conjugation. They give identity potential monodromy on the scalar fibre complex. Translation gives the same assertion at every scalar triple. $\square$

The first nonzero groups for the three target potentials are therefore

n	$\dim_{\mathbb{C}} C_n$	first reduced Milnor degree	first degree of $P_{n,s}$
3	12	11	-6
4	20	23	-8
5	30	39	-10.

Each displayed group has rank one. The calculation concerns ordinary atlas stalks. It does not replace equivariant coefficient complexes by these graded vector spaces.

The rank-two scalar fibre gives a further check on the shifts. Write the traceless parts of A, B, C in the basis $\text{diag}(1, -1), e_{12}, e_{21}$ of $\mathfrak{sl}_2$. If T is the matrix of their three coordinate columns, direct expansion gives $f_2 = 2 \det T$. The scalar directions are free. The determinant-one fibre retracts onto $\text{SU}(3)$ by polar decomposition. The fibration $\text{SU}(2) \rightarrow \text{SU}(3) \rightarrow S^5$ has fibre S^3 . Its rational Serre spectral sequence has nonzero terms only in bidegrees $(0, 0), (0, 3), (5, 0), (5, 3)$, so every differential vanishes. The reduced fibre cohomology has degrees 3, 5, 8, each of rank one. After the normalized shift [8], the scalar coefficient has degrees -4, -2, 1.

1.2 COMPLETE FLAGS DEFINE SCALAR GENERATORS

The rank-one coefficient is $\mathbb{Q}[2]$. Complete flags let these coefficients enter the higher-rank critical coefficient without a choice of a scalar Milnor cycle. Let Fl_n be the variety of complete flags in $\mathbb{C}^n$ and set $p_n = n(n-1)/2$. Let $E_n^{\text{fl}} \subset V_n \times \text{Fl}_n$ classify triples preserving a complete flag. Its matrices are upper triangular in an adapted basis, so $f_n|_{E_n^{\text{fl}}} = 0$. The normal bundle of this incidence has complex rank $3p_n$. Its positive complex Thom class followed by flag integration defines

$$\gamma_n^{\text{fl}} : Rq_{n*} \mathbb{Q}_{E_n^{\text{fl}}} \longrightarrow \mathbb{Q}_{V_n}[4p_n],$$

where q_n forgets the flag. The map is proper because Fl_n is projective.

Pass to quotient stacks, and retain the full critical fibre product

$$\mathcal{T}_n = [E_n^{\text{fl}}/\text{GL}_n] \times_{\mathfrak{X}_n} \mathfrak{M}_n, \quad b_n : \mathcal{T}_n \longrightarrow \mathfrak{M}_n.$$

Thus the off-diagonal commutator equations remain part of the source. Proper exchange for vanishing cycles [4, §3, p. 35] and $\Phi_0 = 1$ give

$$\nu_n : Rb_{n*} \mathbb{Q}_{\mathcal{T}_n}[2n] \longrightarrow \mathcal{P}_n, \quad 2n + 4p_n = 2n^2.$$

The unit of pullback and pushforward then defines degree-zero morphisms

$$h_n : \mathbb{Q}_{\mathfrak{M}_n}[2n] \longrightarrow Rb_{n*} \mathbb{Q}_{\mathcal{T}_n}[2n] \xrightarrow{\nu_n} \mathcal{P}_n, \quad \sigma_n = \frac{h_n}{n!}. \quad (1.2)$$

Every construction commutes with changes of frame. Its normal bundles and flag cycles retain their complex orientations on the action diagram. Thus (1.2) defines morphisms on the quotient stacks.

The sign of h_n can be determined at a matrix A with distinct eigenvalues $\lambda_1, \dots, \lambda_n$. In a transverse eigenbasis chart, the actual summand for $i < j$ is

$$(\lambda_i - \lambda_j)(b_{ij}c_{ji} - b_{ji}c_{ij}).$$

Set $\delta_{ij} = \lambda_i - \lambda_j \neq 0$ and introduce

$$\widehat{c}_{ij} = \delta_{ij}c_{ij}, \quad \widehat{c}_{ji} = \delta_{ij}c_{ji}, \quad q = b_{ij}\widehat{c}_{ji} - b_{ji}\widehat{c}_{ij}.$$

Each rescaling is complex linear with real determinant $|\delta_{ij}|^2 > 0$. Both together have determinant $|\delta_{ij}|^4 > 0$, so they preserve the complex normal orientation. The B -polarized positive disk has normalized coordinates

$$(b_{ij}, b_{ji}, \widehat{c}_{ij}, \widehat{c}_{ji}) = (u, v, -\bar{v}, \bar{u}), \quad q = |u|^2 + |v|^2.$$

Its complex parameter orientation defines a positive support generator. Projection to the normalized lower normal coordinates is $(u, v) \mapsto (v, \bar{u})$, of real degree -1 . In the original coordinates it is $(u, v) \mapsto (v, \delta_{ij}^{-1}\bar{u})$. The additional complex multiplication has positive real determinant $|\delta_{ij}|^{-2}$, so the projection degree remains -1 . The lower A normal cancels the flag tangent by multiplication with the nonzero eigenvalue difference. This complex linear map has positive real determinant. If u_n denotes the product of the B -polarized generators, put

$$e_n = (-1)^{p_n} u_n.$$

All support Thom degrees are even. Their ordered external products introduce no sign. Thus each complete flag contributes e_n , and there are $n!$ complete flags. This proves

$$h_n|_{\text{distinct}} = n!e_n, \quad \sigma_n|_{\text{distinct}} = e_n. \quad (1.3)$$

Here $\mathcal{P}_n = \mathbb{Q}[2n]$ on this locus, since its normal quadratic has $2(n^2 - n)$ variables.

THEOREM 1.2.1 (*Extension of the normalized lowest coefficient*). For $1 \leq n \leq 5$, the stalk of σ_n at any scalar triple induces an isomorphism

$$\mathbb{Q} \xrightarrow{\sim} H^{-2n}(\mathcal{P}_{n,s}).$$

Denote the image of 1 by $v_{n,s}$. On the scalar locus, these classes form a constant, conjugation-equivariant cohomology sheaf. They have identity potential monodromy.

Proof. Positive scaling of all three matrices preserves Z_n and each flag incidence, and multiplies f_n by a positive cube. It preserves the support half-plane, the flag cycles and the Thom orientations. Hence the section of $\mathcal{H}^{-2n}(\mathcal{P}_n)$ induced by h_n is equivariant under positive scaling.

Suppose this section had zero stalk at the origin on the atlas. By the definition of a sheaf stalk, it would vanish on a neighborhood of the origin. Every point of Z_n can be scaled into that neighborhood. Scaling equivariance would force the section to vanish everywhere. This contradicts (1.3). Therefore h_n has a nonzero scalar stalk in degree $-2n$. Proposition 1.1.1 shows that this stalk group is one-dimensional. Division by $n!$ gives the stated isomorphism.

Scalar translations preserve the function, incidences and orientations. They transport this generator to every scalar triple. The morphism already descends under conjugation, so its lowest cohomology sheaf has the trivial scalar stabilizer action. The monodromy assertion also follows from Proposition 1.1.1. $\square$

This theorem specifies the lowest cohomology sheaf. It supplies no splitting of the higher cohomology or of the equivariant derived object.

1.3 THE SCALAR IMAGES OF THE HALL COEFFICIENT MAPS

Fix positive integers d, e with $n = d + e \leq 5$. The product order is submodule first, followed by quotient. Let $E_{d,e}$ classify triples preserving a d -plane, and write

$$\mathcal{F}_{d,e} = [E_{d,e}/\mathrm{GL}_n] \times_{\mathfrak{X}_n} \mathfrak{M}_n, \quad \mathfrak{M}_d \times \mathfrak{M}_e \xleftarrow{a_{d,e}} \mathcal{F}_{d,e} \xrightarrow{b_{d,e}} \mathfrak{M}_n.$$

The diagonal block potential is $f_d + f_e$. The normal Thom map has degree $6de$, and Grassmannian integration has degree $-2de$. Their composite is

$$\gamma_{d,e} : Rq_{d,e*} \mathbb{Q}_{E_{d,e}} \longrightarrow \mathbb{Q}_{V_n} [4de].$$

For analytic functions f, g , the relative-cochain product and the support inclusion

$$\{\mathrm{Re} f \leq 0\} \times \{\mathrm{Re} g \leq 0\} \subset \{\mathrm{Re}(f + g) \leq 0\}$$

define the morphism $\Phi_f K \boxtimes \Phi_g L \rightarrow k^* \Phi_{f+g}(K \boxtimes L)$. The product and support inclusion are those of Massey[3, Lemma 1.2, p. 3, and Proposition 1.4, p. 5]. Here k includes the product of zero fibres in the zero fibre of the sum. This morphism is natural in coefficients and proper direct image: both assertions follow from the same maps of relative cochain complexes. For proper direct image, apply proper base change and the proper Künneth isomorphism[4, §1, pp. 13–14]. Its ordered triple products agree because they use the same product cochains and the same inclusion into the support of $f + g + h$.

Use this support morphism, smooth ambient block pullback, proper exchange, and $\Phi(\gamma_{d,e})$. Restriction to the full critical fibre product defines the actual coefficient map

$$\mu_{d,e} : Rb_{d,e*} a_{d,e}^* (\mathcal{P}_d \boxtimes \mathcal{P}_e) \longrightarrow \mathcal{P}_n. \quad (1.4)$$

Its shift identity is $2d^2 + 2e^2 + 4de = 2n^2$. The critical block map need not be smooth. The construction uses the support comparison as a morphism, including for $f_2 + f_3$. Its invertibility is unnecessary here.

Let $\eta_{d,e}$ denote the unit $\mathbb{Q}_{\mathfrak{M}_n} \rightarrow Rb_{d,e*} \mathbb{Q}_{\mathcal{F}_{d,e}}$. The canonical tensor identification and the maps σ_d, σ_e give a morphism from $\mathbb{Q}_{\mathcal{F}_{d,e}} [2n]$ to the incoming coefficient in (1.4).

THEOREM 1.3.1 (*Scalar lowest-degree Hall images*). For $d + e = n \leq 5$, the following equality holds as morphisms of constructible complexes on $\mathfrak{M}_n$:

$$\mu_{d,e} \circ Rb_{d,e*} a_{d,e}^* (\sigma_d \boxtimes \sigma_e) \circ \eta_{d,e} [2n] = \binom{n}{d} \sigma_n. \quad (1.5)$$

At a scalar triple, the source and target of the induced map in degree $-2n$ are both one-dimensional. In the generators specified by $v_{d,s}, v_{e,s}$ and $v_{n,s}$, this map is multiplication by $\binom{n}{d}$. Its image is the whole lowest-degree target and its kernel in that degree is zero.

Proof. Refine an invariant d -plane by a complete flag in that plane and a complete flag in its quotient. The result is exactly a complete flag in $\mathbb{C}^n$. This identifies the refinement square on the full critical stacks, including its frame changes. Proper base change identifies the corresponding iterated direct images.

The forbidden lower entries for the refined flag are the disjoint union of internal entries in the two blocks and the cross entries. The product of their complex Thom classes is the complete-flag Thom class. Integration over the two internal flag varieties and then the Grassmannian evaluates on the same

complete-flag fundamental cycle. The identity $p_n = p_d + p_e + de$ gives the total degree $4p_n$. All these degrees are even.

On coefficients, proper naturality of the support product commutes the internal $\Phi(\gamma_d^{\text{fl}})$ and $\Phi(\gamma_e^{\text{fl}})$ through the external support morphism. Before the internal pushforwards, all one-dimensional block potentials are zero. Their ordered support product is the constant coefficient on the complete-flag incidence. After the pushforwards, the target potentials remain the actual f_d and f_e . The composite therefore equals h_n when its inputs are h_d, h_e . The adjunction units compose to the complete-flag unit because they send the constant section to the same constant section on the refinement. Dividing by $d!e!$ and using $h_n = n!\sigma_n$ proves (1.5).

At a scalar atlas point, proper base change identifies the incoming stalk with hypercohomology on $\text{Gr}(d, n)$ of the pulled-back coefficient family. Proposition 1.1.1 puts that family in degrees at least $-2d - 2e = -2n$. Theorem 1.2.1 identifies its cohomology sheaf in degree $-2n$ with the constant sheaf generated by $v_d \otimes v_e$. The hypercohomology spectral sequence has no term below this degree. Its degree $-2n$ term is $H^0(\text{Gr}(d, n), \mathbb{Q}) = \mathbb{Q}$. It has no incoming or outgoing differential, since the other relevant terms would have negative sheaf-cohomology degree or lower coefficient degree. Equation (1.5) proves the scalar assertion. $\square$

In particular the first images involving f_3, f_4, f_5 are

(d, e)	degree	incoming lowest class	image
$(2, 1)$	-6	$v_2 \otimes v_1$	$3v_3$
$(2, 2)$	-8	$v_2 \otimes v_2$	$6v_4$
$(3, 1)$	-8	$v_3 \otimes v_1$	$4v_4$
$(4, 1)$	-10	$v_4 \otimes v_1$	$5v_5$
$(2, 3)$	-10	$v_2 \otimes v_3$	$10v_5$

Each incoming class denotes the constant section on the corresponding Grassmannian, with the coefficient morphisms just constructed.

1.4 THREE CONSTITUENTS AND THE REMAINING SCALAR COEFFICIENTS

For an ordered triple (d, e, f) of positive ranks with sum $n \leq 5$, let $\mathcal{F}_{d,e,f}$ be the full critical flag stack with these successive dimensions. Its maps a and b remember the constituents and the total object. The ordered triple support product and the flag Thom map define

$$M_{d,e,f} : Rb_* a^*(\mathcal{P}_d \boxtimes \mathcal{P}_e \boxtimes \mathcal{P}_f) \longrightarrow \mathcal{P}_n.$$

Both parenthesizations use this source, by proper base change. Their ambient Gysin degrees add to $4(de + df + ef)$, since

$$de + (d + e)f = ef + d(e + f).$$

Their normal entries partition the same forbidden entries, and their flag integrations evaluate on the same complex cycle. Ordered support associativity and its coefficient naturality therefore give equality of the two derived maps with $M_{d,e,f}$. This argument retains f_{d+e} and f_{e+f} on the intermediate representation spaces. It introduces no permutation of the inputs.

Refinement to complete flags gives the following restriction of this identity to the constructed coefficient morphisms:

$$M_{d,e,f}(\sigma_d, \sigma_e, \sigma_f) = \frac{n!}{d!e!f!} \sigma_n.$$

The left side includes the adjunction unit, as in (1.5). At a scalar point its source in degree $-2n$ is again one-dimensional, by the same hypercohomology argument on the connected flag variety. For the first three scalar comparisons the two factorizations give

(d, e, f)	left factorization	right factorization	image
$(1, 1, 1)$	$2 \cdot 3$	$2 \cdot 3$	$6v_3$
$(2, 1, 1)$	$3 \cdot 4$	$2 \cdot 6$	$12v_4$
$(2, 2, 1)$	$6 \cdot 5$	$3 \cdot 10$	$30v_5$

These are images of scalar coefficient classes. They follow from the complete-flag morphism and its nonzero scalar restriction.

The lowest coefficient has support on the entire scalar locus $\mathbb{C}^3 \times BGL_n$ and on the distinct-eigenvalue locus. The translation and conjugation compatibilities specify the constant lowest cohomology sheaf on the scalar locus. The ambient derived coefficient remains in every incidence and refinement.

Higher scalar images require further calculations. For example, the rank-two scalar coefficient has degrees $-4, -2, 1$. On a scalar $(2, 1)$ incidence, its degree -2 coefficient tensored with the line coefficient starts in total degree -4 . That degree also receives $H^2(\mathbb{P}^2, \mathbb{Q})$ from the lowest coefficient. The morphism from these contributions to $H^{-4}(P_{3,s})$ is not determined by (1.5). It requires the actual coefficient family, its hypercohomology differentials, and its map to the scalar target. Likewise, the odd rank-two class remains part of the $(2, 2)$ and $(2, 2, 1)$ inputs. Equation (1.1) describes the remaining scalar target groups through commuting-pair homology. It does not identify the Hall maps on those groups.

The constructed scalar images and derived flag identities are rational constructible statements through rank five. A comparison with a holomorphic field-theory carrier requires an additional morphism and compatibility with its operations. Tate structures, higher homotopies and further ranks require their own constructions.

1.5 THE FIRST HIGHER SCALAR HALL MAP

The scalar map for a plane and a line first has two incoming contributions in degree -4 . Let $s = (o, o, o) \in Z_3$, let $G = \text{Gr}(2, 3) = \mathbb{P}^2$, and put

$$D_s = (Rb_{2,1*} a_{2,1}^*(\mathcal{P}_2 \boxtimes \mathcal{P}_1))_s.$$

Stalks use ordinary atlas pullback. Let $\mathcal{K}$ be the pulled-back coefficient family on G , so proper base change gives $D_s = R\Gamma(G, \mathcal{K})$.

PROPOSITION 1.5.1 (*The incoming degree and its proper comparison*). There is a canonical filtration of $H^{-4}(D_s)$ with successive one-dimensional quotients

$$H^2(G, \mathcal{H}^{-6}\mathcal{K}), \quad H^0(G, \mathcal{H}^{-4}\mathcal{K}).$$

In particular $\dim H^{-4}(D_s) = 2$. Define smooth Milnor fibres

$$F = f_3^{-1}(1) \subset V_3, \quad I = \{(A, B, C, H) \in E_{2,1} : f_3(A, B, C) = 1\}, \quad q_1 : I \longrightarrow F.$$

The scalar coefficient map in degree -4 is identified with the following composite:

$$H^{-4}(D_s) \xrightarrow{\sim} H^5(I; \mathbb{Q}) \xrightarrow{q_{1*}} H^{13}(F; \mathbb{Q}) \xrightarrow{\sim} H_{22}^{\text{BM}}(C_3; \mathbb{Q}). \quad (1.6)$$

Here q_{1*} is the proper cohomological Gysin map with complex orientations. The right isomorphism is the scalar support comparison of Proposition 1.1.1.

Proof. The rank-two scalar coefficient has degrees $-4, -2, 1$, each of rank one. The line coefficient has degree -2 . Hence the cohomology sheaves of $\mathcal{K}$ have degrees $-6, -4, -1$ and rank one. They are locally constant: local frames trivialize the tautological plane and quotient, and changes of frames preserve the scalar point. Every local system on $\mathbb{P}^2$ is constant, since $\mathbb{P}^2$ is simply connected.

In the hypercohomology spectral sequence $E_2^{p,q} = H^p(G, \mathcal{H}^q \mathcal{K})$, the terms of total degree -4 are $(p, q) = (2, -6)$ and $(0, -4)$. The first has no outgoing differential because no coefficient occurs below -6 . Its only possible incoming differential has source $(0, -5)$, which is zero. The second has no incoming differential. Its outgoing differential of length two has coefficient degree -5 . The length-three target is $(3, -6)$, where $H^3(\mathbb{P}^2) = 0$. Longer targets have coefficient degree below -6 . Thus both terms survive, giving the filtration and dimension. The filtration does not select a splitting of the two contributions.

Write $E = E_{2,1}$ and $g = f_3|_E$. The variety E is a vector bundle of rank 21 over G , and $\dim_{\mathbb{C}} E = 23$. Its block expression is $g = f_2 + 0$. At potential value one the differential of the rank-two potential is nonzero, so I is smooth of complex dimension 22 . Likewise F is smooth of complex dimension 26 . The restriction q_1 is proper, as a base change of the proper incidence projection.

Positive scaling of the matrix entries contracts E to its zero section G while preserving the sign of $\operatorname{Re} g$. The same radial pair argument as in Proposition 1.1.1 computes the scalar source by

$$D_s \simeq \operatorname{Fib}(R\Gamma(E, \mathbb{Q}) \longrightarrow R\Gamma(I, \mathbb{Q}))[10].$$

The equality uses the rank-one zero-potential comparison, so it retains the actual incoming vanishing-cycle coefficient. Since E retracts to $\mathbb{P}^2$, its cohomology in degrees five and six vanishes. The relative-cochain exact sequence identifies $H^{-4}(D_s)$ with $H^5(I)$. The target is $\operatorname{Fib}(\mathbb{Q} \rightarrow R\Gamma(F, \mathbb{Q}))[18]$. Its degree -4 group is $H^{13}(F)$.

The ambient incidence Gysin map has degree eight. Restriction to the smooth potential fibre is the proper Gysin map q_{1*} , also of degree eight, since $26 - 22 = 4$. The normal Thom construction and flag integration commute with restriction of the potential. Therefore they induce a morphism between the two relative-cochain sequences just used. Both boundary identifications use the same normalized fibre convention. The shift eight is even, so the square has no additional sign. This proves the first two arrows of (1.6). For $n = 3$ and $j = -4$, the Borel–Moore index in (1.1) is $18 - (-4) = 22$, proving the last arrow. $\square$

The projection of I to its preserving pair (A, C, H) has affine fibres over the locus where the diagonal rank-two pair does not commute. Forgetting H from this pair locus is not proper. Fix $H = \langle e_1, e_2 \rangle$ and set

$$A = \operatorname{diag}(0, 1, 2), \quad C_t = te_{12} + e_{13}, \quad B_t = t^{-1}e_{21} \quad (t \neq 0).$$

These matrices preserve H and satisfy

$$[C_t, A] = te_{12} + 2e_{13}, \quad \operatorname{Tr}(B_t[C_t, A]) = 1.$$

The pair (A, C_t) has a limit in U_3 at $t = 0$, since $[C_0, A] = 2e_{13} \neq 0$. Its two diagonal blocks commute at that limit. Thus the limit lies outside the projected incidence locus. The flag remains fixed, so no other limiting flag repairs this missing point. The matrix B_t diverges, consistently with the properness of q_1 before this affine projection.

The image of the two-dimensional group in (1.6) therefore requires the proper Milnor-fibre Gysin map, or a proved replacement retaining its support data. The filtration specifies the two incoming contributions. Their images and the rank of q_{1*} in this degree remain to be determined.

1.6 EVALUATION OF THE DEGREE-MINUS-FOUR IMAGE

The first higher scalar map has rank one. Its kernel has a geometric generator obtained by decorating the internal complete flag. The calculation begins with the rank-two coefficient carried by that decoration.

Let ν_2 be the complete-flag coefficient map. On its scalar fibre $\mathbb{P}^1$, let $h = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$, with $\int_{\mathbb{P}^1} h = 1$. Define

$$w_2 = H^{-2}(\nu_2)(h) \in H^{-2}(P_{2,0}).$$

The input shift of ν_2 is $[4]$, so this notation uses degree two on the flag fibre.

LEMMA 1.6.1 (*The decorated rank-two flag*). The class w_2 is nonzero. Thus it generates $H^{-2}(P_{2,0}) = \mathbb{Q}$.

Proof. In traceless matrix coordinates the potential is $2 \det T$ on $W = \text{Mat}_3(\mathbb{C})$. Scalar matrix coordinates are free. Write $D \subset W$ for the determinant hypersurface. Let $G_0 = \text{Gr}(2, 3) = \mathbb{P}^2$, with tautological plane bundle U , and let

$$\tilde{D} = \text{Hom}(\mathbb{C}^3, U) \longrightarrow D$$

be the proper incidence remembering a plane containing the columns of T . Its total space is smooth of complex dimension eight. Write $z = c_1(\mathcal{O}_{G_0}(1))$. The tautological exact sequence gives $c_1(U) = -z$ and $c_2(U) = z^2$.

Let $D' \subset D$ be the matrices of rank at most one and $E' \subset \tilde{D}$ its inverse image. Their complex dimensions are five and six. Over $D \setminus D'$, the incidence is an isomorphism. The proper resolution of E' that remembers the image line has total space

$$Y = \text{Hom}(\mathbb{C}^3, L) \longrightarrow \text{Fl}(1, 2; 3),$$

where $L \subset U$ is the tautological line. It is a rank-three vector bundle over the complete flag variety. Its map to E' is an isomorphism away from zero matrices. The exceptional locus has complex dimension three, and its image has complex dimension two. Closed-open exact sequences therefore identify $H_{\text{II}}^{\text{BM}}(E')$ with $H_{\text{II}}^{\text{BM}}(Y) = 0$. The last equality follows from the even cohomology of the complete flag variety and the vector-bundle Thom isomorphism.

The top class of E' maps to zero in $H_{12}^{\text{BM}}(\tilde{D})$. To compute it, let $p : \text{Fl}(1, 2; 3) \rightarrow G_0$ forget the line and put $q = c_1(U/L)$. The inclusion of Y into the pullback of $\tilde{D}$ has quotient normal bundle $\text{Hom}(\mathbb{C}^3, U/L)$. Its Euler class is q^3 . Hence the class of E' in the smooth space $\tilde{D}$ is Poincaré dual to

$$p_*(q^3) = c_1(U)^2 - c_2(U) = z^2 - z^2 = 0.$$

Indeed, $q^2 - c_1(U)q + c_2(U) = 0$ by the tautological exact sequence, while integration on the projective-line fibre gives $p_*q = 1$ and $p_*1 = 0$. The resolution $Y \rightarrow E'$ is generically one-to-one, so it pushes its fundamental class to that of E' .

The closed-open sequence for $E' \subset \tilde{D}$ now shows that $H_{12}^{\text{BM}}(\tilde{D}) \rightarrow H_{12}^{\text{BM}}(\tilde{D} \setminus E')$ is an isomorphism. Its kernel is the zero exceptional class and its cokernel maps to $H_{\text{II}}^{\text{BM}}(E') = 0$. Since $\dim_{\mathbb{C}} D' = 5$,

the corresponding map from $H_{12}^{\text{BM}}(D)$ to the open stratum is also an isomorphism. Consequently the proper pushforward is an isomorphism

$$H_{12}^{\text{BM}}(\tilde{D}) = \mathbb{Q} \xrightarrow{\sim} H_{12}^{\text{BM}}(D) = \mathbb{Q}.$$

A line in $\mathbb{C}^2$ determines its traceless stabilizer plane in $\mathfrak{sl}_2$. Under the trace pairing, the perpendicular line consists of nilpotent matrices with that image and kernel. Thus the map from the original flag $\mathbb{P}^1$ to G_0 is the smooth conic $j : \mathbb{P}^1 \hookrightarrow \mathbb{P}^2$. Complex-oriented integration gives

$$j_*1 = 2z, \quad j_*h = z^2.$$

The complete-flag incidence is the restriction of $\tilde{D}$ to this conic. Transitivity of complex Thom maps and proper integration identifies its Gysin map with j_* followed by the universal determinant incidence map.

The support form of this universal map has the following description. Poincaré duality on $\tilde{D}$ takes z^2 to its nonzero degree-twelve Borel–Moore class. Proper pushforward takes this to the nonzero degree-twelve class of D . The closed-open sequence in $\mathcal{W}$, of real dimension eighteen, takes it to the nonzero class in $H^5(\text{GL}_3(\mathbb{C}))$. The support inclusion $D \subset \{\text{Re}(2 \det T) \leq 0\}$ restricts this class to the determinant Milnor fibre. These are the maps of the same relative-cochain sequences defining the normalized coefficient map, so the description also fixes its sign.

The determinant fibration has a product trivialization $\text{GL}_3(\mathbb{C}) \simeq \text{SL}_3(\mathbb{C}) \times \mathbb{C}^*$. Since $H^4(\text{SL}_3) = 0$, restriction induces an isomorphism on H^5 . Multiplication by the positive constant two in the potential preserves the chosen support half-plane. Therefore the image of $j_*h = z^2$ is nonzero in reduced Milnor degree five, hence in normalized scalar degree -2 . This image is exactly w_2 . $\square$

The rank-three target in the required degree also has dimension one. The two possible eigenvalue collisions determine this dimension.

LEMMA 1.6.2 (*The next Borel–Moore group of the commuting-pair variety*). For the variety C_3 of commuting *pairs* of three-by-three matrices,

$$H_{22}^{\text{BM}}(C_3; \mathbb{Q}) = \mathbb{Q}, \quad H^{-4}(P_{3,0}) = \mathbb{Q}.$$

Proof. Remove the locus where A has at most one eigenvalue. The Jordan dimension calculation gives complex dimension at most ten for this locus. Its removal does not change Borel–Moore homology in degrees twenty-two and twenty-three. Denote the remaining space by X , and let $O \subset X$ be the open set where A has three distinct eigenvalues.

The smooth variety O has complex dimension twelve. Up to an equivariantly contractible affine factor, its ordered eigenvalue cover is

$$\text{GL}_3(\mathbb{C})/T \times \text{Conf}_3(\mathbb{C}),$$

where T is the diagonal torus and Conf_3 is the space of ordered distinct triples. The symmetric group permutes both factors, freely on the second. Rational cohomology of O is the invariant part of this cover. The first factor has the cohomology of the complete flag variety. Its H^1 is zero, and its H^2 is generated by three Chern roots with their sum zero. The permutation action has no invariant in this degree.

For the second factor, write

$$(z_1, z_2, z_3) = (a, a + b, a + bc), \quad a \in \mathbb{C}, \quad b \in \mathbb{C}^*, \quad c \in \mathbb{C} \setminus \{0, 1\}.$$

These are global coordinates. Its degree-two cohomology has basis

$$\alpha = d \log b \wedge d \log c, \quad \beta = d \log b \wedge d \log(c - 1),$$

where each logarithmic degree-one class includes the factor $(2\pi i)^{-1}$. The transposition (12) exchanges α and β . The transposition (23) sends $\alpha \mapsto -\alpha$ and $\beta \mapsto \beta - \alpha$. The term $d \log c \wedge d \log(c - 1)$ has zero cohomology class because it comes from the punctured c -line. These two transformations have no common fixed vector. In degree one, the three classes $d \log(z_i - z_j)$ are permuted, so their invariant subspace has dimension one. It follows that

$$H^1(O; \mathbb{Q}) = \mathbb{Q}, \quad H^2(O; \mathbb{Q}) = 0.$$

Poincaré duality gives $H_{23}^{\text{BM}}(O) = \mathbb{Q}$ and $H_{22}^{\text{BM}}(O) = 0$.

The closed complement $D_0 = X \setminus O$ has two irreducible components of complex dimension eleven. On the first, the repeated eigenvalue of A has a Jordan block of size two. On the second, its two-dimensional eigenspace is semisimple. Both strata are irreducible: the centralizer vector bundle over each Jordan stratum gives their parameterization. Their closures are distinct. The generic commuting matrix in the semisimple stratum has two different eigenvalues on that eigenspace. On the Jordan stratum the corresponding restriction is a polynomial in the Jordan block, so its two eigenvalues coincide even in a limit. The opposite containment is excluded by the smaller minimal polynomial on the semisimple stratum. Thus $H_{22}^{\text{BM}}(D_0) = \mathbb{Q}^2$.

The boundary map from $H_{23}^{\text{BM}}(O)$ to these two fundamental classes is nonzero. More precisely, the discriminant of the characteristic polynomial has multiplicities one and two on the two components. At a generic Jordan point, A is cyclic, so the commuting pair variety is locally a centralizer vector bundle over matrix space. The family

$$A_t = \begin{pmatrix} 0 & 1 & 0 \\ t & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

has characteristic discriminant $4t(4 - t)^2$, of order one. At a generic semisimple point, choose the commuting matrix C with three distinct eigenvalues. The local commuting-pair chart then diagonalizes A in its eigenbasis. The family

$$A_t = \text{diag}(t, -t, 2)$$

has discriminant $4t^2(4 - t^2)^2$, of order two. Both are transverse local coordinates in the specified smooth charts. The linking class $d \log \text{disc}(A)$ therefore has boundary a nonzero common multiple of (1, 2), with complex orientations. The closed-open sequence gives

$$H_{22}^{\text{BM}}(X) = \mathbb{Q}^2 / \mathbb{Q}(1, 2) = \mathbb{Q}.$$

Restoring the removed locus preserves this group. Proposition I.1.1 identifies it with $H^{-4}(P_{3,0})$. $\square$

The complete flag in dimension three has successive line bundles L_1, L_2, L_3 . Let $\ell_i = c_1(L_i)$ and define the three degree-zero morphisms

$$\kappa_i : \mathbb{Q}_{\mathcal{M}_3}[4] \longrightarrow Rb_{3*} \mathbb{Q}_{\mathcal{T}_3}[4] \xrightarrow{\ell_i} Rb_{3*} \mathbb{Q}_{\mathcal{T}_3}[6] \xrightarrow{\nu_3} \mathcal{P}_3.$$

The first map is the adjunction unit. Cup product by a first Chern class has cohomological degree two.

LEMMA 1.6.3 (*Three line decorations*). At the scalar atlas point,

$$\kappa_3(1) \neq 0, \quad \kappa_2(1) = 0 \quad \text{in } H^{-4}(P_{3,0}).$$

At the critical point $x = (\text{diag}(0, 0, 1), 0, 0)$, identify the degree -4 coefficient with the rank-two class w_2 tensored with the normalized line class. Then

$$\kappa_1(1) = -2w_2, \quad \kappa_2(1) = 0, \quad \kappa_3(1) = 2w_2.$$

Proof. The spectral projector for the two separated eigenvalue clusters gives a local analytic decomposition into a plane and a line. In this chart, A is block diagonal. Its cross-block commutator operator is invertible. The expression $\text{Tr}(B[C, A])$ therefore splits exactly into the rank-two potential and a non-degenerate bilinear form in the cross-block B, C coordinates. Rescaling the cross-block C coordinates gives two eigenline-pair quadratics. The rescaling is complex linear, so it preserves orientation. Their normalized coefficient, including the free line variables, identifies the target coefficient with $\mathcal{P}_2 \boxtimes \mathcal{P}_1$.

An invariant complete flag separates according to the position of the isolated eigenline. Its fibre at x has three components, each a copy of $\mathbb{P}^1$. On each component, the remaining flag is the complete flag in the double eigenspace. The two cross eigenline pairs contribute $(-1)^2 = 1$ in the B polarization. The conversion to the specified generators also has ratio $(-1)^{p_2+p_1-p_3} = 1$. The flag tangents and lower A normals cancel complex linearly. Thus each component contributes the actual map ν_2 , with coefficient $+1$, to the target identification just given.

Let h be the positive hyperplane class on that component. The tautological subline in the double eigenspace has Chern class $-h$, and its quotient line has Chern class h . The isolated line is constant. Accordingly the three restrictions are

position of isolated line	ℓ_1	ℓ_2	ℓ_3
1	0	$-h$	h
2	$-h$	0	h
3	$-h$	h	0.

Applying $\nu_2(h) = w_2$ and adding the three components proves the values at x .

Each κ_i induces a section of $\mathcal{H}^{-4}(\mathcal{P}_3)$ on the ordinary atlas. These sections are equivariant under positive scaling of the matrices. Since κ_3 is nonzero at x , its origin stalk is nonzero: a section with zero origin germ vanishes near the origin, and scaling would then make it vanish everywhere. Lemma 1.6.2 makes that scalar group one-dimensional. Write $\kappa_2(1) = c\kappa_3(1)$ at the origin. The difference section has zero origin germ, hence vanishes everywhere by the same argument. Its value at x is $-2cw_2$. Lemma 1.6.1 forces $c = 0$. $\square$

The incoming filtration admits a specified geometric lift. Define $\tau_2 : \mathbb{Q}_{\mathfrak{M}_2}[2] \rightarrow \mathcal{P}_2$ by the complete-flag construction with the first Chern class of its quotient line inserted. Its scalar degree -2 class is w_2 . On the scalar Grassmannian $G = \text{Gr}(2, 3)$, let $H = c_1(\mathbb{C}^3/U)$, with $\int_G H^2 = 1$. Define two classes in $H^{-4}(D_s)$ by

$$a = H(v_2 \otimes v_1), \quad b = (Rb_{2,1*} a_{2,1}^*(\tau_2 \boxtimes \sigma_1) \circ \eta_{2,1}[4])_s(1).$$

The first class lies in the degree-two flag contribution. The second is an actual derived coefficient class, not a chosen splitting of the cohomology family.

THEOREM 1.6.4 (*The degree-minus-four scalar image*). The classes a, b form a basis of the two-dimensional scalar source $H^{-4}(D_s)$. Put $w_3 = \frac{1}{2}\kappa_3(1)$ at the scalar point. Then w_3 generates $H^{-4}(P_{3,o})$, and the actual Hall map is

$$H^{-4}(\mu_{2,1})(a) = w_3, \quad H^{-4}(\mu_{2,1})(b) = 0.$$

Thus the degree- -4 image is the whole target and its kernel is $\mathbb{Q}b$. Under the proper Milnor-fibre identification (1.6), the Gysin map has rank one.

Proof. The first class generates the first term of the filtration in Proposition 1.5.1. The edge image of b in its second quotient is $w_2 \otimes v_1$, because the scalar stalk of τ_2 is w_2 at every plane. This is nonzero by Lemma 1.6.1. Hence a, b form a basis.

Refine the plane by its complete flag. Inserting its quotient-line Chern class gives ℓ_2 on the complete flag of dimension three. The proper refinement and support naturality used in Theorem 1.3.1 commute with cup product by this pulled-back Chern class. Consequently $\mu_{2,1}(b) = \kappa_2(1)$. Inserting the outer quotient-line class instead gives ℓ_3 . The normalized plane input is $\sigma_2 = h_2/2$. Thus $\mu_{2,1}(a) = \kappa_3(1)/2 = w_3$. Lemma 1.6.3 proves the displayed values and the nonvanishing of w_3 . The target dimension follows from Lemma 1.6.2. $\square$

The normalization of w_3 is geometric. The section $\kappa_3/2$ specializes to w_2 when a scalar plane is separated from a line. The class w_2 is the positive flag decoration followed by the determinant-incidence support map. This fixes the generator without identifying it with an unspecified Borel–Moore basis.

The computation determines one higher scalar degree for the rank- $(2, 1)$ Hall map. The remaining degrees and the higher scalar images for ranks four and five retain their separate support and extension calculations.

1.7 THE FIRST HIGHER IMAGES THROUGH RANK FIVE

The first higher scalar coefficient for each of f_4 and f_5 is one-dimensional. Complete-flag line classes determine its binary and three-constituent Hall images. All ranks in this section satisfy $2 \leq n \leq 5$. The scalar points, coefficients and orientations remain those of Sections 1.1–1.6.

LEMMA 1.7.1 (*Two adjacent scalar degrees*). At a scalar atlas point,

$$H^{-2n+1}(P_{n,o}) = 0, \quad H^{-2n+2}(P_{n,o}) = \mathbb{Q}.$$

Proof. Set $N = n^2 + n = \dim_{\mathbb{C}} C_n$. Remove the pairs for which $\mathcal{A}$ has at most $n - 2$ distinct eigenvalues. Their complex dimension is at most $N - 2$, by the Jordan calculation. This removal preserves Borel–Moore homology in degrees $2N - 1$ and $2N - 2$. For $n = 2$, the removed locus is empty. Let O be the distinct-eigenvalue locus in the remaining space X . Its ordered cover has, up to a contractible affine factor, the form

$$\mathrm{GL}_n(\mathbb{C})/T_n \times \mathrm{Conf}_n(\mathbb{C}).$$

Here T_n is the diagonal torus. The first factor has $H^1 = 0$. Its H^2 consists of the n Chern roots modulo their sum. Hence its symmetric-group invariants in degree two vanish.

The degree-one cohomology of ordered configurations has basis $a_{ij} = (2\pi i)^{-1} d \log(z_i - z_j)$, for $i < j$, and its cohomology in degree two is generated by their products. Here is a direct induction for the required generation statement. Forgetting the last point is a locally trivial bundle with fibre $\mathbb{C} \setminus \{z_1, \dots, z_{n-1}\}$. Local trivializations follow by moving the points in disjoint disks and extending

those motions to the plane with compact support. The classes $a_{1n}, \dots, a_{n-1,n}$ restrict to the basis dual to the small positive circles in the punctured-plane fibre. These global classes supply permanent cycles for the whole fibre cohomology. The Serre spectral sequence therefore collapses. The cohomology is a free module over the base cohomology on $1, a_{1n}, \dots, a_{n-1,n}$. Induction from $n = 1$ proves the assertion, including the stated degree-one basis.

The symmetric group permutes the unordered-edge basis a_{ij} . It has a one-dimensional invariant subspace in degree one. The exterior square of the full degree-one representation has no invariants. Indeed, for two distinct edges sharing a vertex, the transposition of the other endpoints interchanges the two edges. For disjoint edges, interchanging their two pairs of endpoints has the same effect. Each operation negates the corresponding wedge. Its average over the symmetric group is therefore zero. Since invariants over $\mathbb{Q}$ preserve surjections, the quotient $H^2(\text{Conf}_n)$ also has no invariants. Consequently $H^1(O) = \mathbb{Q}$ and $H^2(O) = 0$.

The complement $X \setminus O$ has two components of complex dimension $N - 1$. They have one double eigenvalue of A , with respectively a Jordan block and a semisimple double eigenspace. All remaining eigenvalues are distinct. Centralizer bundles show that both strata are irreducible. Their closures are distinct by the same commuting-eigenvalue and minimal-polynomial conditions used in Lemma 1.6.2. The discriminant has multiplicities one and two on these components. To check this, use the two-by-two blocks

$$\begin{pmatrix} 0 & 1 \\ t & 0 \end{pmatrix}, \quad \text{diag}(t, -t),$$

and append fixed, pairwise distinct eigenvalues away from zero. All additional factors in the discriminant are units near $t = 0$. The cyclic- A chart in the first case and cyclic- C chart in the second give smooth transverse coordinates, as before.

The closed-open sequence thus has the exact segment

$$0 \longrightarrow H_{2N-1}^{\text{BM}}(X) \longrightarrow \mathbb{Q} \xrightarrow{(1,2)} \mathbb{Q}^2 \longrightarrow H_{2N-2}^{\text{BM}}(X) \longrightarrow 0.$$

The first group is zero and the last is one-dimensional. Restoring the removed locus preserves these groups. The scalar support isomorphism (1.1) gives the two asserted degrees. $\square$

For the complete flag in dimension n , let $\kappa_{n,k} : \mathbb{Q}\mathfrak{M}_n[2n-2] \rightarrow \mathcal{P}_n$ be the complete-flag morphism with $c_1(L_k)$ inserted. Here L_k is its k th successive quotient line. Thus $\kappa_{3,k} = \kappa_k$ and $\kappa_{2,2} = \tau_2$.

PROPOSITION 1.7.2 (*Normalized first higher classes*). The morphism

$$\tau_n = \frac{\kappa_{n,n}}{(n-1)!} : \mathbb{Q}\mathfrak{M}_n[2n-2] \longrightarrow \mathcal{P}_n$$

has a nonzero scalar class w_n in degree $-2n + 2$. It generates the group of Lemma 1.7.1. For each $1 \leq k \leq n$, its scalar line decorations satisfy

$$\kappa_{n,k}(1) = (2k - n - 1)(n - 2)! w_n.$$

These generators agree with w_2 and w_3 already constructed.

Proof. Take a critical triple with one scalar plane and $n - 2$ separated scalar lines. The spectral projectors split its local coefficient into the rank-two coefficient and the line coefficients, together with the nondegenerate cross-block normal quadratic. The same invertible commutator operator and complex rescaling

as in Lemma 1.6.3 give this decomposition. There are $p_n - 1$ cross eigenline pairs. Their B -polarized sign $(-1)^{p_n-1}$ is canceled by the generator-conversion sign $(-1)^{1-p_n}$. Thus every flag component contributes v_2 with coefficient $+1$.

Choose the two positions of the lines in the double eigenspace. The other $n - 2$ labeled lines can occupy the remaining positions in $(n - 2)!$ orders. Each such choice gives one $\mathbb{P}^1$ component of the flag fibre. On it, $c_1(L_k)$ is $-h$ if position k is the first double-eigenspace line, h if it is the second, and zero otherwise. There are $(n - k)(n - 2)!$ negative contributions and $(k - 1)(n - 2)!$ positive contributions. The value of $\kappa_{n,k}$ is therefore $(2k - n - 1)(n - 2)! w_2$ on this stratum, including the line factors.

For $k = n$ this is $(n - 1)! w_2 \neq 0$. Positive-scaling equivariance of the actual cohomology section forces its scalar origin germ to be nonzero. The scalar group is one-dimensional by Lemma 1.7.1. Subtracting the claimed multiple of τ_n from each decoration gives a section whose scalar germ is some multiple of w_n . Its value on the separated-plane stratum determines this multiple, by the same zero-germ scaling argument. This proves the formula and the normalizations. $\square$

Fix $d + e = n \leq 5$ and let $D_{d,e}$ be the actual scalar incoming complex for $\mu_{d,e}$. Let $U \subset \mathbb{C}^n$ be the tautological d -plane on $\text{Gr}(d, n)$ and put $H = c_1(\mathbb{C}^n/U)$. Define the following scalar classes in degree $-2n + 2$:

$$\begin{aligned} A_{d,e} &= H(v_d \otimes v_e), \\ B_{d,e} &= (Rb_* a^*(\tau_d \boxtimes \sigma_e) \circ \eta[2n - 2])_0(1) \quad (d \geq 2), \\ C_{d,e} &= (Rb_* a^*(\sigma_d \boxtimes \tau_e) \circ \eta[2n - 2])_0(1) \quad (e \geq 2). \end{aligned}$$

The maps a, b, η refer to the same full critical correspondence and its adjunction unit as in (1.5). The letters $A_{d,e}, B_{d,e}, C_{d,e}$ denote coefficient classes, not representation matrices.

THEOREM 1.7.3 (Binary first higher images). The displayed classes form a basis of $H^{-2n+2}(D_{d,e})$, whose dimension is $1 + \mathbf{1}_{d \geq 2} + \mathbf{1}_{e \geq 2}$. The actual Hall map has values

$$\begin{aligned} \mu_{d,e}(A_{d,e}) &= \frac{(n - 2)!}{(d - 1)!(e - 1)!} w_n, \\ \mu_{d,e}(B_{d,e}) &= \frac{(d - e - 1)(n - 2)!}{(d - 1)!e!} w_n \quad (d \geq 2), \\ \mu_{d,e}(C_{d,e}) &= \frac{(n - 1)!}{d!(e - 1)!} w_n \quad (e \geq 2). \end{aligned}$$

In particular this map is surjective and has rank one.

Proof. The incoming coefficient has no group one degree above its lowest degree, by Lemma 1.7.1 and $P_1 = \mathbb{Q}[2]$. Its groups two degrees above the lowest come from the first higher class of either nontrivial-rank factor. The Grassmannian has no odd cohomology and has one-dimensional H^2 . Thus the only terms of total degree $-2n + 2$ are its H^2 with the lowest coefficient and its H^0 with these higher coefficients. The same differential argument as in Proposition 1.5.1 makes all these terms permanent. The classes $B_{d,e}, C_{d,e}$ lift their separate nonzero edge classes, whereas $A_{d,e}$ generates the flag contribution. This proves the basis.

Refine the partial flag to a complete flag and commute Chern insertions through the proper support comparison. The three incoming classes become respectively

$$\frac{\sum_{k=d+1}^n c_1(L_k)}{d!e!}, \quad \frac{c_1(L_d)}{(d - 1)!e!}, \quad \frac{c_1(L_n)}{d!(e - 1)!}$$

under the complete-flag coefficient map. Proposition 1.7.2 evaluates them. For the first numerator, $\sum_{k=d+1}^n (2k - n - 1) = de$. The other two numerators are $d - e - 1$ and $n - 1$. These identities give the stated coefficients. The first coefficient is a positive integer, proving surjectivity. $\square$

For the three target ranks the resulting matrices, in the displayed geometric bases and the normalized w_n , are

(d, e)	degree	matrix on (A, B, C) with absent terms omitted
$(2, 1)$	-4	$(1, 0)$
$(2, 2)$	-6	$(2, -1, 3)$
$(3, 1)$	-6	$(1, 1)$
$(4, 1)$	-8	$(1, 2)$
$(2, 3)$	-8	$(3, -2, 6)$.

The same Chern insertions determine the three-constituent comparison. For a scalar flag with dimensions (d, e, f) , set $X = c_1(\mathbb{C}^n/E_d)$ and $Y = c_1(\mathbb{C}^n/E_{d+e})$. Their products with the lowest coefficient give the two flag classes. Include the actual lifts with τ in each constituent of rank at least two. For $(1, 1, 1)$, $(2, 1, 1)$, $(2, 2, 1)$, these form bases by the preceding spectral-sequence argument on the partial flag variety. Its Schubert cells give two independent degree-two classes X, Y and no odd cohomology. The common derived flag map has matrices

(d, e, f)	degree	basis order	image coefficients in w_n
$(1, 1, 1)$	-4	(X, Y)	$(2, 2)$
$(2, 1, 1)$	-6	$(X, Y, \tau_2 \text{ in the first factor})$	$(4, 3, -2)$
$(2, 2, 1)$	-8	$(X, Y, \tau_2 \text{ in the first, second factors})$	$(9, 6, -6, 6)$.

Indeed, complete-flag refinement divides the two flag insertions by $d!e!f!$ and replaces them by the two corresponding suffix sums of the $c_1(L_k)$. A constituent insertion divides instead by the factorial one smaller in that constituent and selects its last line. The values follow from Proposition 1.7.2. The two parenthesisations give these same maps because the ordered support products, Chern pullbacks and complex Thom integrations commute. This determines the first higher scalar images and their stated three-constituent compatibility through rank five.

RESTRICTION TO TWO POINT MODULES

2.1 SPLITTING A TWO-DIMENSIONAL CRITICAL COEFFICIENT

Direct sum gives a restriction from a two-dimensional representation of $\mathbb{C}[x, y, z]$ to two ordered point modules. The vanishing-cycle coefficient supplies the restriction morphism. An Euler class supplies its degree-zero normalization. At coincident points, the coefficient contains an additional torsion summand which the restriction annihilates.

All spaces have their complex analytic topology. Coefficients are rational. For $n = 1, 2$, put

$$V_n = \text{Mat}_n(\mathbb{C})^3, \quad f_n(A, B, C) = \text{Tr}(A[B, C]), \quad Z_n = \{[A, B] = [B, C] = [C, A] = 0\}.$$

The trace pairing identifies Z_n with $\text{Crit}(f_n)$. Write $\mathfrak{X}_n = [V_n/\text{GL}_n]$ and $\mathfrak{M}_n = [Z_n/\text{GL}_n]$. Complexes on these stacks retain the full conjugation descent data. An ordinary atlas pullback introduces no additional shift. Our coefficient and grading are

$$\mathcal{P}_n = \Phi_{f_n} \mathbb{Q}_{\mathfrak{X}_n}[2n^2] \big|_{\mathfrak{M}_n}, \quad \Phi_f K = \text{Cone}(i^* K \longrightarrow \psi_f K)[-1], \quad H^j(K[r]) = H^{j+r}(K).$$

Here i includes the zero fibre and ψ_f denotes nearby cycles. The fibre complex of $u : A \rightarrow B$ has differential $d(a, b) = (da, u(a) - db)$. In particular $\Phi_0 = 1$ and $\mathcal{P}_1 = \mathbb{Q}[2]$. Let

$$\mathcal{H}_n^k = \mathbb{H}^k(\mathfrak{M}_n, \mathcal{P}_n).$$

Only ordinary hypercohomology and algebraic tensor products occur. The function is a submersion away from Z_n , so its vanishing cycles are supported on Z_n . Extension by zero therefore identifies $\mathcal{H}_n$ with ambient vanishing-cycle hypercohomology on $\mathfrak{X}_n$.

Let $T = (\mathbb{C}^*)^2 \subset \text{GL}_2$ be the diagonal torus. The fixed subspace $D = V_2^T$ consists of diagonal triples and is $\mathbb{C}^3 \times \mathbb{C}^3$. Thus $D \subset Z_2$ and

$$d : [D/T] = \mathfrak{M}_1 \times \mathfrak{M}_1 \longrightarrow \mathfrak{M}_2$$

is the ordered direct-sum map. Let L_1, L_2 be the two character lines of T , and put

$$R = H^*(BT; \mathbb{Q}) = \mathbb{Q}[x_1, x_2], \quad x_i = c_1(L_i), \quad |x_i| = 2, \quad \delta = x_2 - x_1.$$

These Chern classes are independent of the positions of the point modules. The contraction of D gives $\mathcal{H}_1 \otimes \mathcal{H}_1 = R[4]$ as graded vector spaces. A polynomial of ordinary degree j has degree $j - 4$ in this tensor product. There is no additional torus acting on the three arrows.

The canonical first map of the vanishing-cycle triangle restricts to

$$\epsilon_D : d^* \mathcal{P}_2 \longrightarrow \mathbb{Q}_{[D/T]}[8]. \tag{2.1}$$

This is a morphism, without an assertion that it is an isomorphism. Equivalently, it is vanishing-cycle pullback for $f_2 \circ d = 0$. The support construction of this pullback is Davison[I, §2.2, equations (19)–(23), pp. 10–11]. The triangle itself defines (2.1) without a comparison theorem. It induces

$$\theta_{1,1} : \mathcal{H}_2^k \longrightarrow R^{k+8} = (\mathcal{H}_1 \otimes \mathcal{H}_1)^{k+4}. \quad (2.2)$$

Its source contains the entire critical coefficient.

2.2 THE EULER FACTOR OF THE EXTENSION CORRESPONDENCE

Let $E \subset V_2 \times \mathbb{P}^1$ classify triples preserving a line. Write $p : [E/\mathrm{GL}_2] \rightarrow \mathfrak{X}_2$ for the proper map forgetting that line. The two diagonal blocks define $q : [E/\mathrm{GL}_2] \rightarrow [D/T]$. The potential f_2 is identically zero on E . The section of q with zero upper entries satisfies $p \circ s = d$.

In a frame with the submodule first, set $W = \mathrm{Hom}(L_2, L_1)$. The ambient extension stack has relative tangent complex

$$[W \xrightarrow{\partial} W^{\oplus 3}], \quad \partial(t) = ((a_2 - a_1)t, (b_2 - b_1)t, (c_2 - c_1)t),$$

where W has degree -1 and $W^{\oplus 3}$ has degree zero. The coordinates (a_i, b_i, c_i) specify the two point modules. Changing the sign of ∂ gives the same quotient action. With the positive complex orientation, the Euler ratio is

$$e_q = \frac{e_T(W^{\oplus 3})}{e_T(W)} = \frac{(-\delta)^3}{-\delta} = \delta^2. \quad (2.3)$$

The calculation concerns the two-term complex on the ambient three-loop representation stack.

The critical incidence is $\mathcal{F} = [E/\mathrm{GL}_2] \times_{\mathfrak{X}_2} \mathfrak{M}_2$. It retains the equations on upper entries. If $v = (a_2 - a_1, b_2 - b_1, c_2 - c_1)$ and u is the upper-entry vector, these equations are $v \wedge u = 0$. For $v \neq 0$, they force $u \in \mathbb{C}v$. For $v = 0$, they impose no condition on $u \in \mathbb{C}^3$. Thus (2.3) comes from the ambient extension complex, whose two vector bundles have constant ranks.

The map q induces an isomorphism on ordinary cohomology with constant coefficients: its upper-entry vector space and unipotent change-of-splitting group are contractible. Consequently $s^* = (q^*)^{-1}$ on this cohomology. Naturality of vanishing-cycle pullback gives

$$\theta_{1,1} = s^* p^* = (q^*)^{-1} p^*,$$

with the shifts in (2.2). Here p^* is pullback from the ambient vanishing-cycle complex to $\Phi_0 \mathbb{Q}_E$, followed by cohomology. Its construction retains the original critical support before applying the coefficient morphism to $\mathbb{Q}_E$. This identifies the map with the rank $(1, 1)$ ambient restriction in the localized Hall construction[2, §3.2.3, Lemma 3.2.4, pp. 26–27].

Definition 2.2.1 (Normalized rank $(1, 1)$ restriction). The localized critical Hall restriction is the degree-zero map

$$\Delta_{1,1}^{\mathrm{loc}} = \delta^{-2} \theta_{1,1} : \mathcal{H}_2 \longrightarrow (\mathcal{H}_1 \otimes \mathcal{H}_1)[\delta^{-1}]. \quad (2.4)$$

Localization is algebraic localization of R at powers of δ . It is not a completion. No coefficient field extension occurs.

The inverse Euler class has degree -4 , which cancels the degree $+4$ of (2.2). The scalar calculation below shows that this particular map has polynomial image.

2.3 THE HALL IDENTITY FOR TWO LINES

The embedding $E \subset V_2 \times \mathbb{P}^1$ has complex codimension three. Its complex Thom map, followed by integration over $\mathbb{P}^1$, is

$$\gamma : R\tilde{p}_*\mathbb{Q}_E \longrightarrow \mathbb{Q}_{V_2}[4], \quad \tilde{p} : E \longrightarrow V_2.$$

All maps descend under GL_2 . Applying Φ_{f_2} , using proper exchange and $f_2|_E = 0$, then restricting to Z_2 , gives

$$\mu_{1,1} : Rb_*\mathbb{Q}_{\mathcal{F}}[4] \longrightarrow \mathcal{P}_2, \quad b : \mathcal{F} \longrightarrow \mathfrak{M}_2.$$

Pullback from the two diagonal blocks and this morphism define $m_{1,1} : \mathcal{H}_1 \otimes \mathcal{H}_1 \rightarrow \mathcal{H}_2$. The order is submodule followed by quotient.

THEOREM 2.3.1 (*Restriction of a rank (1, 1) product*). For $a, b \in \mathbb{Q}[x] = H^*(\mathbb{C}^3 \times B\mathbb{C}^*; \mathbb{Q})$, regarded with shift [2], one has

$$\theta_{1,1}m_{1,1}(a, b) = \delta^2(a(x_1)b(x_2) + a(x_2)b(x_1)), \quad (2.5)$$

$$\Delta_{1,1}^{\mathrm{loc}}m_{1,1}(a, b) = a \otimes b + b \otimes a. \quad (2.6)$$

In particular, if $h_2 = m_{1,1}(1, 1)$ and $\sigma_2 = h_2/2$, then

$$\theta_{1,1}(\sigma_2) = \delta^2, \quad \Delta_{1,1}^{\mathrm{loc}}(\sigma_2) = 1 \otimes 1. \quad (2.7)$$

Proof. Naturality of the first map $\Phi_f K \rightarrow i^* K$ in the coefficient K commutes the triangle with γ . Proper exchange commutes this square with the flag direct image. Since the potential on E is zero, its triangle map is the identity. Thus the composite on the left of (2.5) is the ordinary complex-oriented incidence Gysin map, restricted to D . This comparison is made before taking cohomology.

The two T -fixed flags over D are the coordinate lines L_1, L_2 . At L_1 , the three forbidden lower entries have weight δ . The flag tangent has weight δ . At L_2 , both weights change sign. Their contributions after localization are therefore

$$\frac{\delta^3}{\delta} a(x_1)b(x_2), \quad \frac{(-\delta)^3}{-\delta} a(x_2)b(x_1).$$

Both coefficients are $+\delta^2$. The complex orientation determines these signs. The usual equivariant self-intersection and integration formulas apply to the smooth ambient embedding and the projective flag fibre. Their compatibility with critical pullback is also the one in Davison[I, §2.7, Proposition 2.16, pp. 21–22]. The expression is polynomial, and $R \rightarrow R[\delta^{-1}]$ is injective. Hence the equality already holds in R . Division gives (2.6) and (2.7). The rank-one groups have even degree, so the exchange introduces no Koszul sign. $\square$

There is also a calculation without fixed-point fractions. Let S be the tautological line on $\mathbb{P}(L_1 \oplus L_2)$, write $\xi = c_1(S^*)$ and put $c_1 = x_1 + x_2$, $c_2 = x_1 x_2$. Then

$$\xi^2 + c_1 \xi + c_2 = 0, \quad \int_{\mathbb{P}^1} \xi = 1, \quad c_1(\mathrm{Hom}(S, (L_1 \oplus L_2)/S)) = c_1 + 2\xi.$$

The coefficient of ξ in $(c_1 + 2\xi)^3$ modulo this relation is $2(c_1^2 - 4c_2) = 2\delta^2$. This gives $\theta_{1,1}(h_2) = 2\delta^2$ with the same complex orientation.

2.4 THE FULL SCALAR COEFFICIENT AND ITS RESTRICTION

Let $s = (aI_2, bI_2, cI_2)$ be a scalar triple. Scalar translation preserves the potential, so take $s = 0$. Let P_s be the ordinary atlas stalk of $\mathcal{P}_2$ and define

$$K_s = R\Gamma(BT, P_s).$$

This notation retains the equivariant derived stalk, including its extension data. It does not mean the Borel cohomology of a direct sum of ordinary stalk cohomology groups. Restriction of (2.1) gives

$$\theta_s : H^k(K_s) \longrightarrow R^{k+8}.$$

THEOREM 2.4.1 (*Scalar Euler image and torsion*). The equivariant scalar coefficient has a decomposition of graded R -modules

$$H^*(K_s) = Rv \oplus (R/(\delta^2))w, \quad |v| = -4, \quad |w| = -2, \quad (2.8)$$

where v is the scalar restriction of σ_2 . The map is

$$\theta_s(v) = \delta^2, \quad \theta_s(w) = 0. \quad (2.9)$$

The choice of w depends on a nonzero degree-five Milnor class. The free generator v has the Hall normalization above. The ordinary atlas coefficient has one-dimensional groups in degrees $-4, -2, 1$, and no other groups.

Proof. Write the traceless parts in the ordered basis $H = \text{diag}(1, -1)$, $U = e_{12}$, $L = e_{21}$. If M has the three coordinate columns, direct multiplication gives

$$f_2 = 2 \det M.$$

The three scalar coordinates are free. The compact torus $K = (S^1)^2$ acts on M by left multiplication through $\text{diag}(1, t_1/t_2, t_2/t_1)$.

Choose a K -invariant Hermitian norm. The support description of normalized vanishing cycles uses $P = \{\text{Re } f_2 \leq 0\}$ and the fibre of restriction from a small ball to its complement of P . This description is Massey[3, §1, p. 2]. The radial map $x \mapsto x/(1 + \|x\|/r)$ gives equivariant homotopy equivalences from the global pair to the pair in the radius- r ball. It preserves P , since the potential is homogeneous of degree three. On $H_+ = \{z : \text{Re } z > 0\}$ use the cube root positive on the positive real axis. The equivariant homeomorphism

$$V_2 \setminus P \simeq f_2^{-1}(1) \times H_+, \quad x \mapsto (f_2(x)^{-1/3}x, f_2(x))$$

identifies the two restriction maps. Thus there is an actual fibre sequence

$$K_s \longrightarrow R\Gamma(BK; \mathbb{Q})[8] \longrightarrow R\Gamma(f_2^{-1}(1) \parallel K; \mathbb{Q})[8], \quad (2.10)$$

where $X \parallel K = EK \times_K X$ denotes the Borel construction. The first arrow is θ_s on cohomology.

After constant scaling, the nonzero determinant fibre is $\text{SL}_3(\mathbb{C})$. Polar decomposition $M = UP$ gives the deformation $UP \mapsto UP^t$ onto $\text{SU}(3)$. It is equivariant for the displayed unitary left action. The scalar-coordinate factor is equivariantly contractible. Split $K = K_c \times K_e$ by $(z, t) \mapsto (zt, z)$. The central factor K_c acts trivially, while

$$K_e \longrightarrow \text{SU}(3), \quad t \mapsto \text{diag}(1, t, t^{-1})$$

acts freely by left multiplication. Consequently the last Borel space in (2.10) has the homotopy type

$$BK_e \times Y, \quad Y = K_e \setminus \mathrm{SU}(3).$$

The map taking the first row of a unitary matrix gives a bundle $\mathrm{SU}(2) \rightarrow \mathrm{SU}(3) \rightarrow S^5$. Its structure group acts by left multiplication in the lower 2×2 block, which contains K_e . Taking the left K_e quotient gives

$$S^2 = K_e \setminus \mathrm{SU}(2) \longrightarrow Y \longrightarrow S^5. \quad (2.11)$$

Local orthonormal completion of a first row supplies local trivializations. Changes of completion induce right translations on $K_e \setminus \mathrm{SU}(2)$, preserving orientation. The rational Serre spectral sequence of (2.11) has terms only in bidegrees $(0, 0)$, $(0, 2)$, $(5, 0)$, $(5, 2)$. Every differential vanishes for degree reasons.

Put $u = c_1(t) = x_1 - x_2 = -\delta$. The principal circle bundle $\mathrm{SU}(3) \rightarrow Y$ restricts on the displayed S^2 fibre to $S^3 \rightarrow S^2$. In $\mathrm{SU}(2)$ coordinates $\begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$, left multiplication by $\mathrm{diag}(t, t^{-1})$ sends (a, b) to (ta, tb) . Thus its quotient is the Hopf map $S^3 \rightarrow \mathbb{P}^1$, whose transition function on the equator has degree one up to orientation. Thus u restricts to a generator of $H^2(S^2; \mathbb{Q})$, up to its orientation sign. Let $\beta \in H^5(Y; \mathbb{Q})$ be the pullback of a sphere generator. Fibre integration shows $u\beta \neq 0$. There is no degree-four cohomology, and odd-degree squares vanish over $\mathbb{Q}$. Therefore

$$H^*(Y; \mathbb{Q}) = \mathbb{Q}[u]/(u^2) \otimes \Lambda(\beta), \quad |\beta| = 5.$$

The homomorphism induced by $Y \rightarrow BK_e$ sends u to this class. After adjoining the central variable this gives

$$H^*(f_2^{-1}(1) \parallel K; \mathbb{Q}) = R/(\delta^2) \otimes \Lambda(\beta). \quad (2.12)$$

The map from R is the quotient into the even summand.

The long exact sequence of (2.10) now gives

$$0 \longrightarrow (R/(\delta^2)) w \longrightarrow H^*(K_s) \xrightarrow{\theta_s} \delta^2 R[8] \longrightarrow 0, \quad |w| = 5 + 1 - 8 = -2.$$

The quotient is a free graded R -module with generator in degree -4 . Equation (2.7), restricted to the scalar diagonal, supplies a section v whose image is δ^2 . This proves (2.8) and (2.9).

For the ordinary atlas stalk, the bundle $S^3 \rightarrow \mathrm{SU}(3) \rightarrow S^5$ gives $H^*(\mathrm{SU}(3); \mathbb{Q}) = \Lambda(\alpha_3, \alpha_5)$. Its spectral sequence has no possible differential. The product of the two generators evaluates nontrivially on the total orientation by fibre integration. The reduced cohomology degrees are 3, 5, 8. The normalized fibre and shift [8] send these degrees to $-4, -2, 1$. In degree -4 , the equivariant-to-ordinary edge map is an isomorphism: there is no lower stalk cohomology which could contribute or receive a differential in that total degree. Hence v restricts to a nonzero ordinary lowest class. $\square$

The scalar Borel spectral sequence gives a direct description of the extension responsible for the torsion. Normalize α_3 so that

$$d_4 \alpha_3 = \delta^2, \quad d_4(\alpha_3 \alpha_5) = \delta^2 \alpha_5.$$

The first differential is nonzero because (2.12) has no degree-four class in the effective-circle variable. Its sign fixes the chosen transgression normalization. The second equality is the product rule. Thus the ordinary odd scalar class participates in the equivariant coefficient, although $H^*(K_s)$ is concentrated in

even degrees. Taking the tensor product of (2.8) with $R/(x_1, x_2)$ does not compute the ordinary atlas stalk. The original fibre complex in (2.10) retains that distinction.

Potential monodromy also retains its ambient meaning. The maps $(A, B, C) \mapsto (A, e^{i\vartheta} B, C)$ preserve centered balls and multiply the potential by $e^{i\vartheta}$. At $\vartheta = 2\pi$ the map is the identity. These maps commute with conjugation and the constant-to-nearby-cycle map. They give identity monodromy on the scalar equivariant fibre complex. The triangle morphisms commute with this monodromy.

2.5 POLYNOMIAL IMAGE AND SPECIALIZATION

COROLLARY 2.5.1 (*Cancellation in rank two*). The map (2.4) has image in $\mathcal{H}_1 \otimes \mathcal{H}_1$. It therefore defines a unique degree-zero map

$$\Delta_{1,1} : \mathcal{H}_2 \longrightarrow \mathcal{H}_1 \otimes \mathcal{H}_1, \quad \theta_{1,1} = \delta^2 \Delta_{1,1}.$$

It satisfies (2.6) and (2.7). Its scalar counterpart sends v to $1 \otimes 1$ and annihilates $(R/(\delta^2))w$.

Proof. Include one scalar point in D and retain its T stabilizer. Restriction $H^*([D/T], \mathbb{Q}) \rightarrow H^*(BT, \mathbb{Q})$ is the identity on R , since D is equivariantly contractible. Naturality of (2.1) identifies the scalar restriction of $\theta_{1,1}(c)$ with θ_s of the restriction of c . Theorem 2.4.1 puts this image in $\delta^2 R$. Therefore $\theta_{1,1}(c)$ is divisible by δ^2 . Multiplication by δ^2 is injective on R . This proves existence and uniqueness of the polynomial map. The scalar statements follow from (2.9). $\square$

Collision of the two point positions means restriction along the scalar diagonal $\mathbb{C}^3 \rightarrow D$. This leaves x_1, x_2 and δ as stabilizer classes. The natural coefficient square for this restriction is the square used in the corollary. It pulls back the ambient vanishing cycles rather than replacing them by vanishing cycles of a zero function on the scalar locus.

Forgetting stabilizers is a different operation. It sends x_1, x_2 to zero. There is no ring homomorphism $R[\delta^{-1}] \rightarrow \mathbb{Q}$ extending that specialization. After the proved cancellation, evaluation of the polynomial output is well defined. It sends the scalar Hall generator v to 1. The raw ordinary stalk morphism $P_s \rightarrow \mathbb{Q}[8]$ is zero: its target degree is -8 , whereas the source degrees are $-4, -2, 1$. Consequently evaluation of the cancelled cohomological map and ordinary pullback of the raw coefficient morphism are distinct operations.

Interchange of L_1, L_2 sends δ to $-\delta$ and preserves δ^2 . The two direct-sum maps differ by conjugation in GL_2 . Naturality therefore makes $\Delta_{1,1}$ symmetric under this interchange. Its scalar generator is the complete-flag class $h_2/2$. This is also the input normalization for products involving two rank-two constituents. For a flag of type $(2, 2, 1)$, the restriction constructed here can be applied to either rank-two input and sends its lowest class to the ordered pair of rank-one classes. A coproduct identity for the rank-five output would additionally require restriction maps with rank-five source and a comparison of the corresponding flag correspondences. The map above establishes the rank $(1, 1)$ identity without that additional assertion.

RESTRICTION TO A PLANE AND A LINE

The first higher scalar class survives restriction to a plane and a line. Its image contains the discriminant torsion of the plane coefficient. This torsion disappears after the plane splits into two lines. The two restrictions therefore require different localizations.

The potentials, normalized vanishing cycles, complex orientations, and ordinary atlas pullbacks are those of Section 1.1. All cohomology has rational coefficients and the analytic topology. The groups below retain their full Borel coefficient complexes. Every localization is algebraic. No completion occurs.

3.1 THE SCALAR SUPPORT MAP

Let $G = \mathrm{GL}_3(\mathbb{C})$ and $L = \mathrm{GL}_2(\mathbb{C}) \times \mathbb{C}^* \subset G$. Write U for the defining plane and Q for the defining line. Their ordered direct sum gives

$$j : V_2 \times V_1 \longrightarrow V_3, \quad f_3 j = f_2 + \mathfrak{o}.$$

Put $K_{3,G} = R\Gamma(BG, P_{3,\mathfrak{o}})$ and $K_{2,L} = R\Gamma(BL, P_{2,\mathfrak{o}} \otimes \mathbb{Q}[2])$. The notation for $K_{2,L}$ includes the line coefficient. Define

$$R = H^*(BL; \mathbb{Q}) = \mathbb{Q}[c_1, c_2, y], \quad c_i = c_i(U), \quad y = c_1(Q), \quad |c_i| = 2i, \quad |y| = 2.$$

Scalar translation identifies every coincident scalar point with the origin.

For a compact subgroup $K \subset L$, write $X \parallel K = EK \times_K X$. Choose the product Hilbert–Schmidt norm on all matrix entries. It is invariant under unitary conjugation and phase rotation of B . Let $F_n = f_n^{-1}(\mathfrak{i})$ and $P_n^{\leq} = \{\mathrm{Re} f_n \leq \mathfrak{o}\}$. The support realization of normalized vanishing cycles gives

$$\begin{aligned} K_{3,L} &\simeq \mathrm{Fib}(C^*(BL) \longrightarrow C^*(F_3 \parallel L)) [18], \\ K_{2,L} &\simeq \mathrm{Fib}(C^*(BL) \longrightarrow C^*((F_2 \times V_1) \parallel L)) [10]. \end{aligned} \tag{3.1}$$

Here Borel constructions for L use its maximal compact subgroup. Here $K_{3,L} = R\Gamma(BL, P_{3,\mathfrak{o}}|_L)$ restricts the equivariant coefficient before taking L -equivariant cohomology.

Indeed, positive radial contraction identifies the global support pair with every sufficiently small centered pair. The cube root on $\{\mathrm{Re} z > \mathfrak{o}\}$ trivializes the complement as the potential-one fibre times this half-plane. Both maps commute with the compact group and the block inclusion. This proves (3.1) as fibre complexes, including their maps. The support construction is Massey[3, §1, p. 2].

Restriction of these pairs, preceded by restriction from G to L , defines the actual degree-eight map

$$\theta_{2,1} : H^k(K_{3,G}) \longrightarrow H^{k+8}(K_{2,L}). \tag{3.2}$$

Its map on Milnor complements is the pullback under $F_2 \times V_1 \hookrightarrow F_3$. Equivalently, it is the scalar restriction of

$$j^* \mathcal{P}_3 \longrightarrow (\mathcal{P}_2 \boxtimes \mathcal{P}_1) [8].$$

Vanishing-cycle pullback and the zero-function Thom–Sebastiani identification construct this morphism before cohomology. See Davison[I, §2.2, equations (19)–(23), pp. 10–11]. Thus the source of (3.2) contains the entire scalar critical coefficient, including its equivariant extension data.

3.2 THE EULER DENOMINATOR AND THE TARGET TORSION

Let $E_{2,1} \subset V_3 \times \text{Gr}(2, 3)$ be the incidence of triples preserving a plane. Its projection $p : [E_{2,1}/G] \rightarrow [V_3/G]$ is proper. The block projection $q : [E_{2,1}/G] \rightarrow [(V_2 \times V_1)/L]$ has the split section s , and $ps = j$. Its relative tangent complex is

$$[W \longrightarrow W^{\oplus 3}], \quad W = \text{Hom}(Q, U),$$

in degrees $-1, 0$. For block matrices A_U, A_Q , and similarly B, C , its differential is

$$t \longmapsto (A_U t - t A_Q, B_U t - t B_Q, C_U t - t C_Q),$$

up to a common sign. The potential on the incidence is $q^*(f_2 + 0)$. Smooth pullback of vanishing cycles and the contractible upper-entry and unipotent fibres identify q^* with an isomorphism on cohomology. Its inverse is s^* . This is an ambient incidence statement. The critical fibre product retains all upper-entry commutator equations.

Set

$$d = c_1^2 - 4c_2, \quad r = y^2 - c_1 y + c_2, \quad e = r^2.$$

If x_1, x_2 are the Chern roots of U , then $d = (x_2 - x_1)^2$ and $r = (y - x_1)(y - x_2)$. The complex-oriented Euler ratio of the relative tangent is $e_L(W^{\oplus 3})/e_L(W) = r^2$. It has cohomological degree eight. The normalized scalar restriction is therefore

$$\Delta_{2,1}^{\text{loc}} = e^{-1} \theta_{2,1} : H^*(K_{3,G}) \longrightarrow H^*(K_{2,L})[e^{-1}]. \quad (3.3)$$

It has degree zero. The identity $s^* p^* = (q^*)^{-1} p^*$ identifies it with the localized Hall restriction of Jindal–Kaubrys–Latyntsev[2, §3.2.3, Lemma 3.2.4, pp. 26–27] in these ranks and conventions.

Let v denote the scalar class of $\sigma_2 \boxtimes \sigma_1$. Let t denote the class of $\tau_2 \boxtimes \sigma_1$, where τ_2 inserts the quotient-line Chern class in the plane flag. Put $w = t - c_1 v$.

PROPOSITION 3.2.1 (*The plane coefficient after adjoining a line*). There is a decomposition of graded R -modules

$$H^*(K_{2,L}) = Rv \oplus (R/(d))w, \quad |v| = -6, \quad |w| = -4. \quad (3.4)$$

Multiplication by e is injective on this module. In particular, w survives localization at e . The normalized restriction of the plane to two lines sends v to $1 \otimes 1 \otimes 1$ and w to zero.

Proof. First retain the diagonal torus in GL_2 . Theorem 2.4.1 computes its actual fibre complex and gives one free class and one class annihilated by $(x_2 - x_1)^2$. The flag formula of Theorem 2.3.1, with the quotient line decorated, gives

$$\theta_{1,1}(t) = d(x_1 + x_2), \quad \theta_{1,1}(v) = d.$$

Hence $w = t - c_1 v$ belongs to the torsion summand. Lemma 1.6.1 identifies the ordinary image of t with the nonzero positive flag decoration w_2 . The ordinary image of $c_1 v$ is zero. Thus w is the nonzero generator of that summand in its lowest degree.

Restriction from GL_2 to its torus identifies rational equivariant cohomology with Weyl invariants in the torus cohomology. This statement also holds with the present equivariant complex: the flag fibration and Leray–Hirsch use the tautological line class and the same pulled-back coefficient. The classes v, t, w descend under GL_2 . Taking invariants is exact over $\mathbb{Q}$. The invariant coefficient ring is $\mathbb{Q}[c_1, c_2]$, and the invariant torsion quotient is $\mathbb{Q}[c_1, c_2]/(d)$. Adjoining γ and the line shift $[2]$ proves (3.4).

The free summand has no e -torsion. In the other summand, $c_2 = c_1^2/4$ and

$$e \bmod d = (\gamma - c_1/2)^4 \neq 0 \quad \text{in } R/(d) = \mathbb{Q}[c_1, \gamma].$$

This quotient is a domain, so multiplication by e is injective there. The plane restriction follows from its two displayed images. $\square$

The ordinary plane-and-line coefficient has degrees $-6, -4, -1$. The last degree comes from the equivariant extension in (2.10). Tensoring the cohomology module (3.4) with $R/(c_1, c_2, \gamma)$ loses that class. Forgetting stabilizers therefore acts on the original derived coefficient. There is no augmentation $R[e^{-1}] \rightarrow \mathbb{Q}$ extending $c_1, c_2, \gamma \mapsto 0$. The raw ordinary scalar pullback is zero on cohomology in every degree. Indeed, the source vanishes below degree -6 , whereas the shifted target $(P_{2,0} \otimes \mathbb{Q}[2])[8]$ has degrees $-14, -12, -9$. Thus polynomial evaluation of selected normalized images is a separate operation from forgetting stabilizers in the raw support map.

3.3 RESTRICTION OF THE COMPLETE-FLAG MORPHISM

Write $\nu_n(p)$ for the complete-flag coefficient morphism with a polynomial p in the successive line Chern classes inserted. Its scalar equivariant value has the shifts of ν_n and the cohomological degree of p . For the internal plane flag, put $s = c_1(S)$ and $u = c_1(U/S)$. Then

$$s + u = c_1, \quad su = c_2, \quad \nu_2(1) = 2v, \quad \nu_2(u) = c_1v + w. \quad (3.5)$$

The notation on the right includes the external line coefficient.

LEMMA 3.3.1 (*The three positions of the external line*). For every polynomial p in three ordered line Chern classes,

$$\theta_{2,1}\nu_3(p) = e\{\nu_2(p(\gamma, s, u)) + \nu_2(p(s, \gamma, u)) + \nu_2(p(s, u, \gamma))\}. \quad (3.6)$$

The equality holds in $H^*(K_{2,L})$, including its torsion summand.

Proof. Use the central circle in L which acts trivially on U and by scalar multiplication on Q . Its fixed complete flags have three components, each $\mathbb{P}(U)$. They are indexed by the position of Q in the complete flag. Only the two cross-block pairs move. The internal plane flag remains.

Localization at r suffices for this fixed-locus calculation. Here is an explicit concentration argument on the flag variety. Besides the three fixed components, the L -orbits have representatives

$$\begin{aligned} (\langle e_1 \rangle &\subset \langle e_1, e_2 + q \rangle), \\ (\langle e_1 + q \rangle &\subset \langle e_1, q \rangle), \\ (\langle e_1 + q \rangle &\subset \langle e_1 + q, e_2 \rangle), \end{aligned}$$

where e_1, e_2 span U and q spans Q . Projection to U, Q and the intersection of the plane with U give this list directly. On the first stabilizer, $U/\langle e_1 \rangle \simeq Q$. On the other two, $\langle e_1 \rangle \simeq Q$. Consequently $\mathrm{Hom}(Q, U)$

has a trivial line subquotient, so its top Chern class r restricts to zero on every such stabilizer. The finite orbit filtration and its relative-cohomology exact sequences kill the complement after inverting r . This argument applies to equivariant coefficient complexes and to their pullbacks to the incidence. Over a fixed flag, a nonzero moving matrix entry gives a nowhere-zero section of $\text{Hom}(Q, U)$ or its dual. Its Euler class is again r and vanishes on that locus. The six possible cross-block maps give a finite open cover of the remaining nonfixed incidence. Mayer–Vietoris gives the same concentration.

The complex Thom self-intersection formula now expresses the proper incidence map as the sum of its three fixed contributions. One can apply it directly to the support pairs defining Φ : the Thom maps, excision maps, and proper direct images commute with restriction to $\{\text{Re } f \leq 0\}$. Thus the coefficient on each fixed component is the actual ν_2 . It is not the constant coefficient on the block critical locus. This compatibility is also the construction in Davison[I, §2.7, Proposition 2.16, pp. 21–22].

For each cross pair, let λ be the tangent weight which moves the earlier line towards the later line. The three forbidden lower entries each have weight λ . Their Thom Euler class divided by the flag tangent Euler class is $\lambda^3/\lambda = \lambda^2$. The two cross pairs therefore contribute $r^2 = e$, independent of the position of Q . Every orientation is complex, and all Thom degrees are even. The three restrictions of p are the three displayed substitutions. This proves (3.6) after inverting r . Inverting r and $e = r^2$ gives the same ring. Proposition 3.2.1 makes localization of the target injective, so the equality holds before localization. $\square$

Put $C = c_1 + y$, the restriction of the first Chern class of the rank-three defining representation. The polynomial identity (3.5) gives

$$\nu_2(s) = c_1 v - w.$$

Inserting 1 and each of the three line classes in (3.6) therefore gives

class	$\Delta_{2,1}^{\text{loc}}(\text{class})$
σ_3	v
$\kappa_{3,1}$	$2Cv - 2w$
$\kappa_{3,2}$	$2Cv$
$\kappa_{3,3} = 2\tau_3$	$2Cv + 2w.$

(3.7)

For example, the last line is $\nu_2(u) + \nu_2(u) + y\nu_2(1) = 2Cv + 2w$. The normalization $\sigma_3 = \nu_3(1)/6$ gives the first line.

THEOREM 3.3.2 (*The first two scalar generators*). Let $z_3 = \tau_3 - C_1\sigma_3 \in H^{-4}(K_{3,G})$, where $C_1 \in H^2(BG)$ is the first Chern class. Then

$$\Delta_{2,1}^{\text{loc}}(\sigma_3) = v, \quad \Delta_{2,1}^{\text{loc}}(z_3) = w \neq 0. \quad (3.8)$$

The restriction has polynomial image on the $H^*(BG)$ -submodule generated by σ_3 and z_3 . The second image has annihilator (d) in R and survives cross-block localization.

Proof. Divide (3.6) by e and use (3.7). The map is linear over $H^*(BG)$ through block restriction. Equation (3.4) proves the annihilator assertion. $\square$

Polynomiality in this theorem concerns the specified submodule. Divisibility of $\theta_{2,1}(H^*(K_{3,G}))$ by e in the full module (3.4) is a further question. The map (3.3) is defined on the full source regardless of that divisibility.

3.4 COMPATIBILITY WITH THE ACTUAL HALL IMAGE

The scalar incoming coefficient for a plane and a line is

$$D_o = (Rb_{2,1*}a_{2,1}^*(\mathcal{P}_2 \boxtimes \mathcal{P}_1))_o.$$

Its G -equivariant version retains the tautological plane and quotient on $\text{Gr}(2, 3)$. Let a^G be its lowest class multiplied by $c_1(Q)$. Let b^G use the actual morphism τ_2 in the plane factor. These are equivariant lifts of the geometric classes a, b in Theorem 1.6.4.

Refining the plane by its complete flag gives identities of coefficient morphisms, before taking scalar cohomology:

$$\mu_{2,1}(a^G) = \frac{1}{2}\nu_3(\ell_3) = \tau_3, \quad \mu_{2,1}(b^G) = \nu_3(\ell_2) = \kappa_{3,2}. \quad (3.9)$$

The factor $1/2$ is the normalization $\sigma_2 = \nu_2(1)/2$. Proper refinement, cup product, and the complex Thom maps commute on the same support pairs. Consequently (3.7) computes the composite on these actual incoming classes:

$$\begin{aligned} \Delta_{2,1}^{\text{loc}} \mu_{2,1}(a^G) &= Cv + w, \\ \Delta_{2,1}^{\text{loc}} \mu_{2,1}(b^G) &= 2Cv. \end{aligned} \quad (3.10)$$

There is also an exact equivariant identity behind the zero ordinary image of b :

$$\kappa_{3,2} = 2C_1\sigma_3 \quad \text{in } H^{-4}(K_{3,G}). \quad (3.11)$$

To prove it, use the scalar degrees $H^{-6}(P_{3,o}) = \mathbb{Q}$, $H^{-5}(P_{3,o}) = 0$, and $H^{-4}(P_{3,o}) = \mathbb{Q}$. They were computed from the commuting-pair strata in Proposition 1.1.1 and Lemma 1.7.1. The Borel spectral sequence in total degree -4 has only $H^2(BG)H^{-6}(P_{3,o})$ and $H^0(BG)H^{-4}(P_{3,o})$. Neither supports a differential: the intervening scalar group and odd base cohomology vanish, and lower scalar groups are zero. Thus its dimension is two. The images Cv and $Cv + w$ of $C_1\sigma_3$ and τ_3 are linearly independent over $\mathbb{Q}$. The map is therefore injective in this degree. Equation (3.7) proves (3.11).

Forgetting stabilizers in (3.9) and (3.11) gives

$$\mu_{2,1}(a) = w_3, \quad \mu_{2,1}(b) = 0.$$

These are the actual ordinary degree -4 images, with the same positive flag normalizations as Theorem 1.6.4. Equation (3.10) is their equivariant refinement. Only its polynomial outputs can be evaluated at $c_1 = c_2 = y = 0$. They evaluate to w and zero, respectively. In this degree w has the ordinary image $w_2 \otimes v_1$. This evaluation does not define an augmentation of (3.3) on its full localized target.

The proper map for the multiplication remains

$$q_1 : I = E_{2,1} \cap f_3^{-1}(1) \longrightarrow F_3.$$

The varieties I, F_3 are smooth of complex dimensions 22, 26. Their Gysin map has degree eight. Together with the support boundary maps, it is the map $H^5(I) \rightarrow H^{13}(F_3)$ of Proposition 1.5.1. The restriction uses the separate closed inclusion $F_2 \times V_1 \hookrightarrow F_3$ and cohomological pullback. Thus (3.10) compares a proper Gysin map with a specified pullback of the same support coefficient. The affine projection to noncommuting pairs is unnecessary. Its projected incidence map is nonproper, as the family following Proposition 1.5.1 shows.

3.5 THE FIRST COASSOCIATIVE FLAG

Let $T_3 = (\mathbb{C}^*)^3$, with ordered Chern roots x_1, x_2, x_3 , and put

$$R_3^T = \mathbb{Q}[x_1, x_2, x_3], \quad D = \prod_{1 \leq i < j \leq 3} (x_j - x_i)^2.$$

The direct inclusion of three scalar lines defines a raw support map

$$\theta_{1,1,1} : H^k(K_{3,G}) \longrightarrow (R_3^T)^{k+18}.$$

It is the first arrow of the vanishing-cycle triangle after restriction to the diagonal triples. Define $\Delta_{1,1,1}^{\text{loc}} = D^{-1}\theta_{1,1,1}$, with target $R_3^T[D^{-1}][6]$ and degree zero.

THEOREM 3.5.1 (*Three-line coassociativity*). As maps on the full scalar source $H^*(K_{3,G})$,

$$(\Delta_{1,1}^{\text{loc}} \otimes 1)\Delta_{2,1}^{\text{loc}} = \Delta_{1,1,1}^{\text{loc}} = (1 \otimes \Delta_{1,1}^{\text{loc}})\Delta_{1,2}^{\text{loc}}. \quad (3.12)$$

Both iterated maps take values in $R_3^T[D^{-1}][6]$. Here $\Delta_{1,2}^{\text{loc}}$ uses the reversed block order and its corresponding Euler factor. On σ_3, τ_3, z_3 , the common values are respectively 1, $x_1 + x_2 + x_3$, and 0.

Proof. The two composites of block inclusions are the same ordered diagonal inclusion $V_1^3 \hookrightarrow V_3$. Pullbacks of the support pairs compose with this identity. The zero-function Thom–Sebastiani maps are the unit identifications, so the two raw composites are $\theta_{1,1,1}$. All inserted shifts are even and give no Koszul sign. This is the scalar support realization of the coassociativity in Jindal–Kaubrys–Latyntsev[2, Lemma 2.2.2, pp. 19–20].

The two products of Euler denominators are equal:

$$\begin{aligned} (x_2 - x_1)^2 ((x_3 - x_1)(x_3 - x_2))^2 &= D, \\ (x_3 - x_2)^2 ((x_2 - x_1)(x_3 - x_1))^2 &= D. \end{aligned}$$

They have degree twelve, the total raw shift. Each constituent Euler factor becomes invertible in the displayed target ring, so both composites are defined there. This proves (3.12). Proposition 3.2.1 sends v to 1 and w to zero. Apply it to (3.8) and $\Delta_{2,1}^{\text{loc}}(\tau_3) = Cv + w$. $\square$

For the complete scalar flag, let $X = \ell_2 + \ell_3$ and $Y = \ell_3$. These are the two degree-two quotient Chern classes on $\text{Fl}(1, 2; 3)$. Its actual coefficient map sends them to $\kappa_{3,2} + \kappa_{3,3}$ and $\kappa_{3,3}$. Their plane-and-line restrictions are

$$4Cv + 2w, \quad 2Cv + 2w.$$

The ordinary Hall images are both $2w_3$. The common three-line restrictions are $4(x_1 + x_2 + x_3)$ and $2(x_1 + x_2 + x_3)$, respectively. This computes the first higher three-constituent flag while preserving the equivariant terms which vanish on the ordinary atlas.

The construction establishes the scalar rank- $(2, 1)$ restriction and the stated three-line compatibility. A full polynomial restriction requires divisibility on every scalar degree. A comparison with holomorphic Chern–Simons observables requires a separate morphism preserving these coefficients and correspondences.

BIBLIOGRAPHY

- [1] Ben Davison, *The critical CoHA of a quiver with potential*, The Quarterly Journal of Mathematics **68** (2017), no. 2, 635–703. arXiv:1311.7172v5; locators refer to the preprint.
- [2] Shivang Jindal, Šarūnas Kaubrys, and Alexei Latyntsev, *Critical CoHAs, vertex coalgebras and deformed Drinfeld coproducts*, 2026. arXiv:2603.21707v1, Lemma 2.2.2 and Lemma 3.2.4.
- [3] David B. Massey, *The Sebastiani–Thom isomorphism in the derived category*, 1999. arXiv:math/9908101v1, §1, pp. 2–5.
- [4] ———, *Notes on perverse sheaves and vanishing cycles*, 2025. arXiv:math/9908107v13, §1, pp. 13–14; §3, p. 35.